INCH-POUND
MIL-PRF-10304/13E
w/AMENDMENT 3
18 March 2020
SUPERSEDING
MIL-PRF-10304/13E
w/AMENDMENT 2
28 August 2013

PERFORMANCE SPECIFICATION SHEET

# METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: VOLTMETER, AC (FLUSH MOUNTING, ROUND FLANGE, 3.5 INCH), STYLE 36

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-10304.

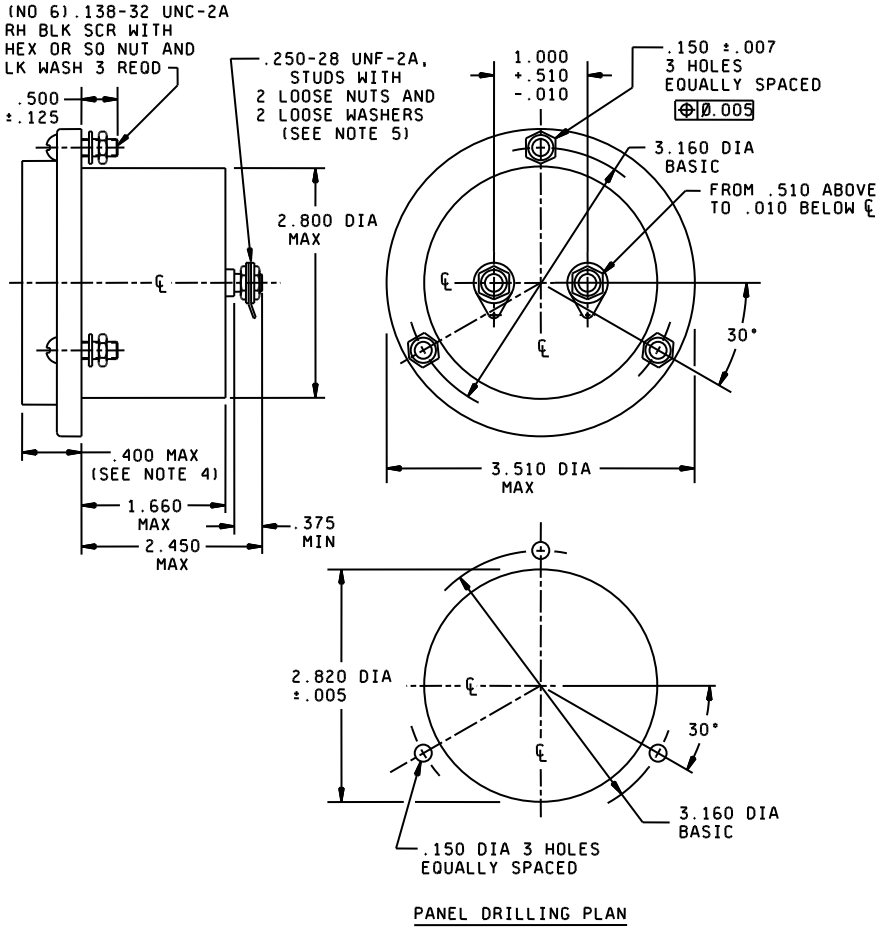



FIGURE 1. Meter, panel type.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .005 | .13 | .250 | 6.35 | 1.660 | 42.16 |
| .007 | .18 | .375 | 9.53 | 2.450 | 62.23 |
| .010 | .25 | .400 | 10.16 | 2.800 | 71.12 |
| .125 | 3.18 | .500 | 12.70 | 2.820 | 71.63 |
| .138 | 3.51 | .510 | 12.95 | 3.160 | 80.26 |
| .150 | 3.81 | 1.000 | 25.40 | 3.510 | 89.15 |

NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 inch for two place (.XX), and ± .005 inch for three place (.XXX) dimensions.
2. Tolerance is ± 0.5 (½) degree on angles.
3. Metric equivalents are given for general information only.
4. Maximum projection from front panel including mounting gaskets (if used).
5. Other forms of terminals, as specified in MIL-PRF-10304, are permitted. The specified minimum length applies to stud-type terminals only. Terminals shall be equally spaced, from each side of vertical centerline.

FIGURE 1. Meter, panel type - Continued.

REQUIREMENTS:
(Readings expressed as a percent of full-scale value.)

Dimensions and configuration: See figure 1.

Full-scale ranges: See table I.

Scale:
Length: 2 inches minimum.
Pointer deflection: 90 degrees minimum.

Position influence: ± 2 percent change of 60 degrees rotation from normal operating position.

Accuracy: ± 2 percent.

Overshoot: 67 percent maximum.

Response time: 2.5 seconds maximum.

Power consumption (loss) (at end scale deflection): 4 volt-amperes maximum.

Frequency range (± 2 percent change):

| Nominal frequency | Extreme range |
|---|---|
| 60-Hz voltmeters (AC) | 25-125 Hz |
| 400-Hz voltmeters (AF) | 360-440 Hz |
| 800-Hz voltmeters (AE) | 720-880 Hz |

High temperature cycling:
± 3 percent.
± 2 percent permanent change.

Temperature influence: ± 0.75 percent change.

Exposure to extreme temperatures: ± 3 percent permanent change.



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Overload capacity:
Sustained overload:
± 1 percent temporary zero shift.
± 1 percent permanent zero shift.
± 2 percent permanent change.

Dielectric withstanding voltage: 3,000 volts rms.

TABLE I. Full-scale ranges.

| Volts (AC, AE, AF) | | | |
|---|---|---|---|
| 1.5 | 15 | 80 | 300 |
| 3 | 30 | 150 | 1/ 500 |
| 8 | 50 | | 1/ 800 |
| 10 | | | 1/ 1000 |

1/ Supplied with external shunt.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. This document references MIL-PRF-10304.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
DLA – CC

Preparing activity:
DLA – CC

(Project 6625-2020-007)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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